

- **Tween:** If anything other than default data type and value is required for interpolation, then Tween can be used. Output double by an Animatable is not required. The current value is provided simply with a mapping function. The current value can be found using the value method.
- **Tween.animate:** Call `animate()` to use Tween object on the Tween. It passes on the `ControllerObject`.
- **Animation Notifications:** Listeners and Status Listeners can be defined with `addStatusListener()` and `addListener()`. With a change in animation value, a listener is called. Listener calls `setState()` for a rebuild. Every time the animation status changes a `StatusListener` is called.

Animation Examples

We have listed out some animation examples-

1. **Rendering Animations:** Rendering with an Animation object needs the object to be stored as a member of the widget. Then its value can be used while deciding how to draw. A `vsync` object needs to be passed while defining an Animation Controller. Every time a new number gets generated by Animation, the `build()` is forced to be called again because the current frame gets marked dirty.

The Dart's cascade notation means, along with the return value from `animate()`, `addListener()` method is called. The notation is represented as `..addListener()`.

2. **Simplifying with AnimatedWidget:** The separation of the core object from the animation code is done using `AnimatedWidget` base class. The state object does not need to be maintained by the Animated Widget in order to hold the animation. The current value of the animation is used by `AnimatedLogo` while drawing itself. The following animated logo class should be added:-

- `AnimationController` and `Tween` are still managed by the `LogoApp`. The Animation object is passed to the `AnimationLogo`.
- In order to make an animated widget, the `AnimatedWidget` helper class can be used.
- An `AnimatedWidget` can be used to create a widget that presents a reusable animation.
- `AnimatedBuilder` helps in separating the transition from the widget.
- The `AnimatedWidgets` examples in Flutter API are :- `SizeTransition`, `ScaleTransition`, `SlideTransition`, `PositionedTransition`, `FadeTransition`, `DecoratedBoxTransition`, `AnimatedModalBarrier`